Elliptic Functions and Integrals

This document describes how elliptic integrals and functions are used in the C47/R47 calculator.

In some cases, elliptic function parameters are used differently depending on which book or software package you are looking at. This is a matter of slight differences in terminology and notation across the elliptic function literature. Especially notable is the modulus parameter, interpreted differently and with the letter k or m, with different meaning. When comparing computations from one place or another, the answers may appear to disagree, but only because the parameter was used differently.

This document attempts to explain those differences so that you can get consistent results. If parameters are mapped properly from one usage to another, the answers should come out well.

Numerical results are given to 5 digits to the right of the decimal point where possible, for consistency in comparisons. The text [1] uses that precision, which is easily set on the C47/R47 with f DISP, FIX, 5. Every screen shown in this note was captured after that one setting, so the display groups the five decimals as `0.198 72` rather than running them together.

Two of the functions here take an angle, and they read it in whatever angular mode the calculator is in. The worked examples in the body are in radians, so set f MODE, RAD before following them; the comparison tables in Section 5 are the exception and are in degrees, which is the factory setting. Getting this wrong does not produce an error, only a wrong answer, so it is worth checking the small **r** in the status bar before you start.

The elliptic functions live in one menu of their own. Open the extended-functions menu with g X.FN and page to its second page, where the first key on the blue row is g Ellipt. The menu runs to three pages and the paging wraps, so ↑ gets there in one press where ↓ would take you to the third page instead. Everything this note uses is on the single page that `Ellipt` opens, and the keystrokes below name each function by the label the screen prints above its soft key. Figure 2 on page 12 is that page as the calculator draws it.

> **WARNING**
> The elliptic functions are a build option, and on the DM42 and DM42n only firmware package 2 carries them. Packages 1, 3 and 4 leave `Ellipt` and every function in it out, and the menu labels are struck through where they would have been. If `Ellipt` is struck out on your calculator, you are running a package without the elliptic option rather than doing anything wrong.

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REV	DATE	DESCRIPTION
1.0	2026-04-17	First published, against firmware 0.109.03.02, with Annex 1, a worked tutorial by Jaco Mostert, appended to the published PDF.
1.1	2026-07-30	Moved to the shared design language in `sources/style/c47appnote.sty`, and the date pinned rather than taken from the build clock. Keystrokes are drawn in the house notation and every one of them was replayed against the simulator: that added the seven screen captures and corrected three steps, the menu paging key, the repeat of `MVAR`, and the angular mode the ϕ examples need. Every numerical result in the note was reproduced unchanged.

1 Basics of Elliptical Functions and Integrals

What are elliptic functions and integrals?

In some ways they are just solutions to differential equations that have no apparent bearing on ellipses at all.

Here we will explore those issues and show simple examples of computations with elliptic functions that you can try on a C47 or R47 calculator, or in Mathematica.

1.1 Review: Basic Geometry of an Ellipse

Many people studied conic sections in high school. See [3] for example. A conic section is a geometrical idea from ancient Greek mathematics. Certain curves can be represented as a conic section geometrically combined with analytical (Cartesian) geometry. Table 1 gives the examples.

Table 1 Examples of conic sections. The ellipse is the one case with an eccentricity that is neither fixed nor unbounded, and that eccentricity is the modulus the elliptic functions take.

NAME	EQUATION	ECCENTRICITY, e
Circle	$x^2 + y^2 = r^2$	0
Parabola	$y^2 = 4ax$	1
Ellipse	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	$\sqrt{1 - \frac{b^2}{a^2}}$
Hyperbola	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$\sqrt{1 + \frac{b^2}{a^2}}$

Since we are interested in ellipses here, let us note that the ellipse has eccentricity $e \in [0, 1)$. In the special case where an ellipse has eccentricity 0, it is actually a circle.

Besides the equation given above for an ellipse, it can also be represented parametrically, with ϕ the angle drawn in Figure 1:

$$x = a\cos\theta,\ \ y = b\sin\theta,\ \ \theta \in [0, 2\pi)$$

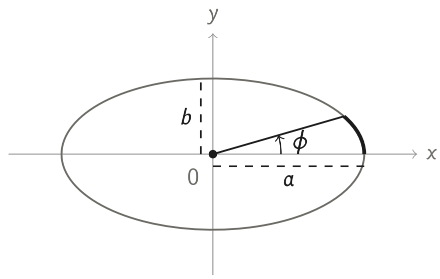


Figure 1 The ellipse $x^2/a^2 + y^2/b^2 = 1$ with semi-axes a and b. The heavy arc is the piece swept from the x axis out to the parametric angle ϕ, and its length is what the incomplete elliptic integral of the second kind returns.

2 A Problem to Solve with an Ellipse: Arc Length and Inverse

One sees online comments, e.g. [4], about elliptic integrals and functions deriving from arc length problems, but actual examples are hard to find. Most of the applications for elliptic integrals and functions go back to problems from specialized differential equations or engineering problems.

I persisted in this question and found answers that tie back to the functions in the C47/R47 calculator. Here are my findings.

First, there is the problem of notation and parameter names. This is discussed in [5]. An elliptic function like $sn(u, m)$ can be described with either m or k given as the second argument, and sometimes the second argument is not mentioned at all.

Similarly, an elliptic integral can be described as $K(m)$ or $F(\phi, k)$ or $F(\phi, m)$, with apparently k and m being different things. And then, their relationship to geometric parameters that we know from simple conic descriptions, like a or b or e, is not obvious.

The simple answer is, that when you see k as a parameter, it is called the "modulus", but actually it is equal to the eccentricity e as seen in elementary conic section terminology. Some sources instead refer to m as the modulus.

However, though k is often favoured in mathematical treatments, you often see in software such as the C47/R47 or Mathematica a parameter called m. Actually, this is almost the same thing: $m = k^2 = e^2$. This m parameter is just the square of eccentricity.

Now, a typical second year calculus textbook for engineering students teaches how to calculate arc length in two- or three- dimensional Euclidean space, using integration. The Olmsted book, [2], does a good job of this.

To paraphrase [2]: If C is an arc with parametrization functions $x = f(t), y = g(t)$, with continuous derivatives on $[a, b]$, then C is rectifiable with length

$$L = \int_a^b \sqrt{(f'(t))^2 + (g'(t))^2}\, dt$$

As an example of this, for the ellipse described above, we get (with a little rearrangement):

The total arc length of the ellipse is [2]

$$4a \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2\theta}\, d\theta$$

This leads to the following. Incomplete elliptic integral of the second kind:

$$E(\phi, \mathbf{m}) = \int_0^{\phi} \sqrt{1 - \mathbf{m} \sin^2 \theta} \, d\theta$$

Complete elliptic integral of the second kind:

$$E(\mathbf{m}) = \int_0^{\pi/2} \sqrt{1 - \mathbf{m} \sin^2 \theta} \, d\theta$$

Confusingly, in different parts of the literature, you will see this version instead. Incomplete elliptic integral of the second kind:

$$E(\phi, \mathbf{k}) = \int_0^{\phi} \sqrt{1 - \mathbf{k}^2 \sin^2 \theta} \, d\theta$$

Complete elliptic integral of the second kind:

$$E(\mathbf{k}) = \int_0^{\pi/2} \sqrt{1 - \mathbf{k}^2 \sin^2 \theta} \, d\theta$$

See also [8].

Just understand that k is eccentricity of the ellipse, $m = k^2$, and that for both you have the restriction $k \in [0, 1)$, $m \in [0, 1)$.

So now we have an understandable application if you followed all the bits about conic sections, eccentricity, and arc length. We simply want to take an ellipse and calculate its arc length. The incomplete elliptic integral of the second kind does this for arbitrary angle ϕ, meaning we start at angle 0 and rotate counter-clockwise to ϕ, measuring arc length as we go. The complete elliptic integral of the second kind calculates the perimeter of the ellipse.

Let us take an example and run it through both Mathematica and the C47/R47.

Take $a = 5$, $b = 1$, so we have the ellipse $\frac{x^2}{5^2} + \frac{y^2}{1} = 1$.

We have eccentricity $k = e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{24/25}$. Both Mathematica and the C47/R47 use a parameter named m, so we use modulus $m = .96$.

> **TIP**
> The calculator will do that step for you. g a,b→m on the blue row of `Ellipt` takes the two semi-axes off the stack and returns the modulus directly, so the m for this ellipse is 5 ENTER 1 g a,b→m. It sorts the two axes itself, so the order you enter them in does not matter.

In Mathematica, we can compute this:

```
EllipticE[.96]

--> 1.05050
```

As a sanity check, we can also compute:

```
Integrate[Sqrt[1 - .96 (Sin[t])^2], {t, 0, Pi/2}]

--> 1.05050
```

Also:

```
ArcLength[{5 Sin[t], Cos[t]}, {t, 0, Pi/2}]

--> 5.25251
```

which is 5 times the value above, also confirming our correctness.

If we want to calculate the partial arc length, say for an angle ϕ of .2 radians and the same modulus, we run

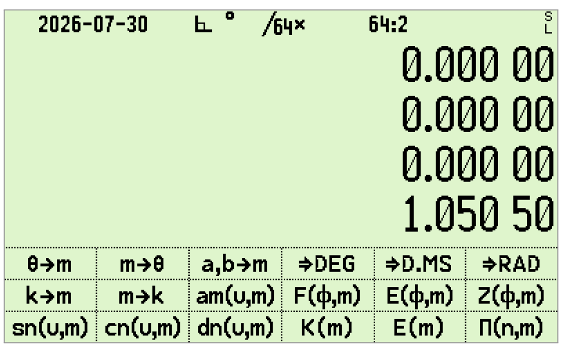
```
EllipticE[.2, .96]

--> 0.19872
```

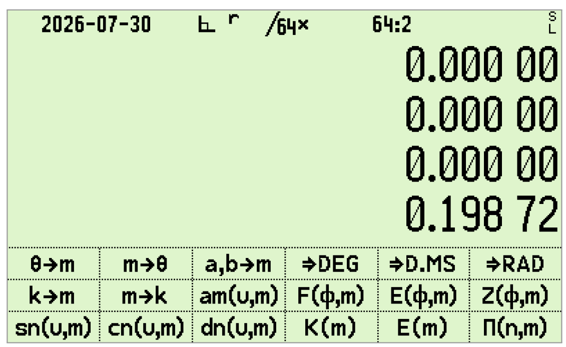
On the C47/R47, we do this:

[.][9][6] [E(m)] → `X: 1.050 50`

For the partial angle, in radians:

[.][9][6] [ENTER] [.][2] [f] [E(ϕ,m)] → `X: 0.198 72`





Left, the perimeter parameter: [E(m)] at $m = .96$ leaves `1.050 50`, matching Mathematica. Right, the partial arc at $\phi = .2$, which needs radian mode: the **r** in the status bar has replaced the degree sign, and the answer is `0.198 72`. The menu stays open under the result, so the next function is one key away.

This confirms that elliptic integrals of the second kind give correct arc length calculations.

> **NOTE**
> The complete integral is on the menu's first row and the incomplete one is on the orange row above it, so [E(m)] is a plain soft key and [f] [E(ϕ,m)] takes a shift first. The same split holds for the first kind, [K(m)] against [f] [F(ϕ,m)]. The complete forms take one argument and the incomplete forms take two, m in **Y** and the angle in **X**.

Now let us look at the inverse problem. For a given modulus and a given arc length, we want to know what angle ϕ produces that arc length.

Unfortunately, my researches show that there is no such function in either the mathematical literature or the software packages of Mathematica or C47/R47. We have to resort to other methods such as root solving. See [7] for more on this issue.

In Mathematica, try this:

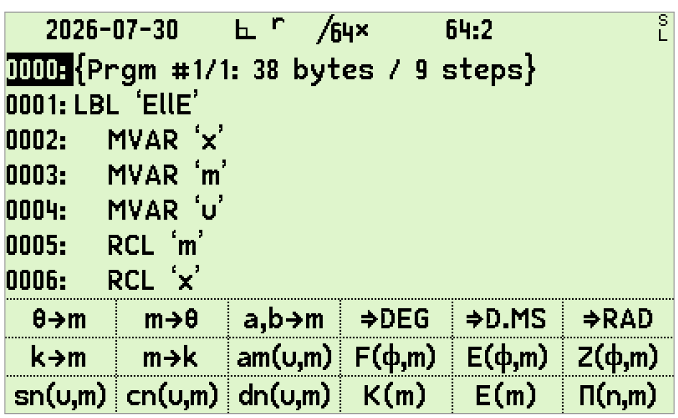
```
FindRoot[EllipticE[x, .96] == .19872, {x, .5}]

--> 0.20000
```

Correct answer.

In the C47/R47, enter a program:

```
0001: LBL 'EllE'
0002:   MVAR 'x'
0003:   MVAR 'm'
0004:   MVAR 'u'
0005:   RCL 'm'
0006:   RCL 'x'
0007:   E(ϕ,m)
0008:   RCL- 'u'
0009: END
```

To key it in, open a new program with XEQ . . and switch to program-entry mode with f PRGM. Start the label with g LBL, then f α, type E l l E, and ENTER. Declare the first variable next: P.FN1, then P.FN2, then MVAR, then f α x ENTER. The menu stays on P.FN2 afterwards, so the second and third variables are just MVAR f α m ENTER and MVAR f α u ENTER. Then key the body: RCL f α m ENTER, RCL f α x ENTER, then g X.FN ↑ to reach the second page of the extended-functions menu, g Ellipt, f E(ϕ,m), and finally RCL – f α u ENTER. Leave program-entry mode with f PRGM. Application Note 29 [9] walks through program entry from the beginning if any of that is unfamiliar.

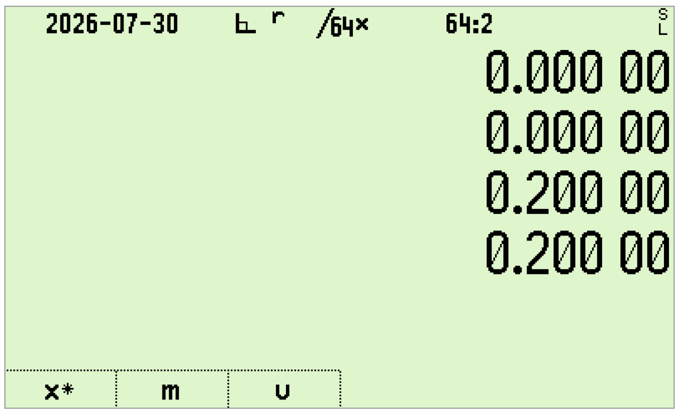


EllE keyed in, seen from the top: nine steps in 38 bytes. The listing continues past RCL x with f E(ϕ,m) and RCL– u, and the Ellipt menu is still open underneath because the last function was taken from it. The three MVAR declarations are what let the solver ask for m and u and solve for x.

> **NOTE**
> `MVAR` is not on `f` `ADV`, which is where most readers look for it. In program-entry mode it is two pages into the program-functions menu already on the screen: `P.FN1` on soft key 6, `P.FN2` on soft key 6 again, then `MVAR`. Declaring a variable with `MVAR` is what lets the solver ask you for it afterwards.

If you run this program `EllE` under `f` `ADV`, `SOLVE` with a modulus of .96 and a u value of .19872, you will get an answer of 0.20000. Naming the program puts its three declared variables on the soft keys in the order they were declared, so `m` takes the modulus, `u` takes the arc length, and `x` is the one to solve for:

`.` `9` `6` `m` `.` `1` `9` `8` `7` `2` `u` `x` → `x: 0.200 00`



The angle recovered, `0.200 00`, with the star on the `x` soft key marking the variable that was solved for. Run this one in degrees by mistake and it returns `11.458 99`, which is the same answer in the other unit and no error at all.

This accounts for $E(m)$ and $E(\phi, m)$ on the C47/R47, which are the Elliptic Integrals of the Second Kind.

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3 Elliptic Integrals of the First Kind

The integrals of the first kind are $K(m)$ (complete) and $F(\phi, m)$ (incomplete).

Equations:

$$F(\phi, m) = \int_0^{\phi} \frac{d\theta}{\sqrt{1 - m\sin^2\theta}}$$

$$K(m) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - m\sin^2\theta}}$$

These also have a geometric interpretation, from Complex Analysis. On constructing a conformal map from an ellipse to a unit circle, the elliptic integral of the first kind tells you the angular displacement needed to map arc length in the ellipse to the corresponding point in the circle [6].

In this case, there is an inverse mapping. $am(x)$, described in Section 5, is the inverse of the incomplete elliptic integral of the first kind.

4 Elliptic Integrals of the Third Kind and Jacobi Zeta

$\Pi(n, m)$ is the incomplete Elliptic Integral of the third kind. It has applications in Physics, e.g. electrostatic potential.

In Mathematica, use `EllipticPi[n,m]`. On the C47/R47, use [Π(n,m)].

Comparison of values from Mathematica and the C47/R47:

```
EllipticPi[0.4,0.6]

--> 2.59092
```

[.][6] [ENTER] [.][4] [Π(n,m)] → `X: 2.590 92`

The Jacobi Zeta function is found in Mathematica as `JacobiZeta[ϕ,m]` and on the C47/R47 as [f] [Z(ϕ,m)]. It has applications in physics and in interpretation of elliptic arc compared with circle.

Mathematica example:

```
JacobiZeta[2,.5]

--> -0.11777
```

C47/R47 example:

[.][5] [ENTER] [2] [f] [Z(ϕ,m)] → `X: -0.117 77`

These two match.

5 Jacobi Elliptic Functions

As a foundation to these functions, there are first the four Theta functions. These arise as solutions to a partial differential equation related to heat diffusion over a slab:

$$\kappa\nabla^2\theta = \frac{d\theta}{dt}$$

These give rise to functions $\theta_1(z, q)$, $\theta_2(z, q)$, $\theta_3(z, q)$, $\theta_4(z, q)$.

The C47/R47 calculator does not implement these Theta functions, but Mathematica does: `EllipticTheta[a,u,q]`. These are used for the theory behind the following Elliptic functions, which also have applications in the Elliptic integrals.

$sn(u, m)$, $cn(u, m)$, $dn(u, m)$. These are rendered in Mathematica as

```
JacobiSN[u,m]
JacobiCN[u,m]
JacobiDN[u,m]
```

These functions have some interesting identities. For example:

$$sn(u,m)^2 + cn(u,m)^2 = 1$$

$sn(u,0) = \sin(u)$ for u in radians.

$cn(u,0) = \cos(u)$ for u in radians.

We can test these. The sn is unaffected by the angular mode, but the [SIN] it is being compared against is not, so this one needs radian mode, f [MODE], [RAD]:

[0] [ENTER] [.][2] [sn(u,m)] [.][2] [SIN]

puts matching values in the **X** and **Y** registers:

```
Y:  0.19867
X:  0.19867
```

As an example of understanding how to use parameters, the book [1] shows in Table C that for $k = 0.5$ and $x = 0.8$, $sn(x,k) = 0.70421$.

However, this calculation on the C47/R47:

[.][5] [ENTER] [.][8] [sn(u,m)] → `X:  0.690 93`

produces the number .69093: an apparent mismatch.

Is this wrong? NO. The calculator got a different answer from the book, but both the calculator and the book are right. WHAT?

Yes, explaining this issue clearly is one of the main motivations for writing this document. The answer is in the difference in parameter interpretation. At the risk of being too verbose, I **must** explain this point.

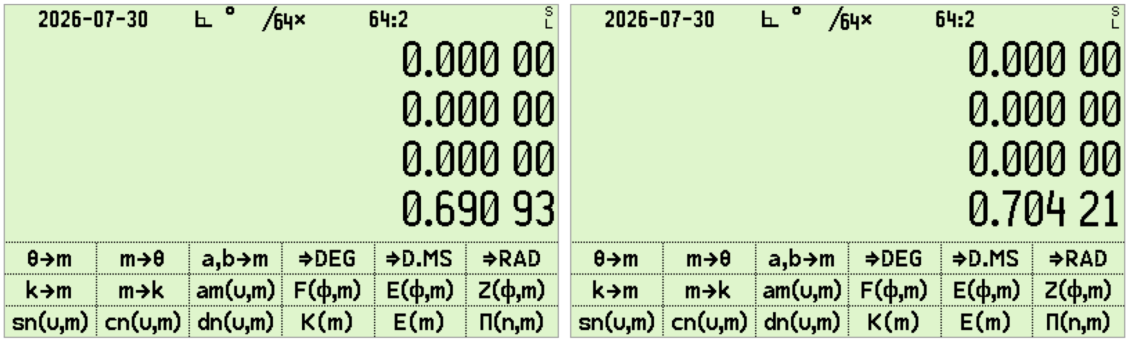
Earlier in this document, I used a lot of space to explain the k parameter versus the m parameter. Some books or software packages expect k as the modulus parameter, and some expect m as the modulus parameter. As explained earlier, $m = k^2$.

Where the book's table has $sn(u,k)$ with $k = 0.5$, the calculator wants to run $sn(u,m)$ and we have to square the k parameter to get the m parameter.

Knowing that $m = k^2$, we can recalculate on the C47/R47 this way:

[.][5] f [x^2] [.][8] [sn(u,m)] → `X:  0.704 21`

and get the matching value of 0.70421.



The whole of the note in two screens. Left, the book's $k = 0.5$ handed straight to sn(u,m), which reads it as m and returns `0.690 93`. Right, the same k squared first, which is the m the function was asking for, returning the book's `0.704 21`. Neither screen is wrong; they answer two different questions.

> **TIP**
> f k→m on the orange row of `Ellipt` is that squaring step, named for what it is, and f m→k takes you back. Reaching for it instead of f x^2 says in the keystroke itself which convention you are converting from, which is worth something in a program you will read again in a year.

If this sounds strange, think of trig functions. The result of the sine function depends on whether the parameter is interpreted as being in degrees or radians or other units. Accordingly, you might have to do a conversion between degrees and radians in some cases to make the sine function give the expected result.

Here, the result of $sn()$ depends on whether the second parameter is interpreted as the eccentricity or the square of the eccentricity of the ellipse. Accordingly, you might have to square a parameter to make it compatible with the function you are calling.

Table 2 gives more examples comparing $sn()$ calculations with tables in the back of the [1] textbook. The Jacobi functions take an argument rather than an angle, so these do not depend on the angular mode; the elliptic integrals in Table 3 do, and are in degrees.

Table 2 Lawden comparison examples, Jacobi functions. The first three result columns are the book's, at modulus k; the last three are the calculator's, at modulus $m = k^2$. They agree once the parameter is converted.

k	x	$sn\ x$	$cn\ x$	$dn\ x$	$sn(u, m)$	$cn(u, m)$	$dn(u, m)$
.5	.8	.70421	.70999	.93596	.70421	.70999	.93596
.6	1.3	.93416	.35686	.82816	.93416	.35686	.82816

Incomplete elliptic integrals using degrees are in Table 3.

Table 3 Lawden comparison examples, incomplete elliptic integrals. Lawden's F and D against the calculator's F and E, with the angle in degrees.

k	$\phi°$	LAWDEN $F(\phi, k)$	$F(\phi, k)$	LAWDEN $D(\phi, k)$	$E(\phi, k)$
.4	36	.63459	.63459	.62215	.62215
.6	65	1.21431	1.21431	1.06311	1.06311

Apart from this issue for $sn()$, there is also the Jacobi amplitude function, represented in the C47/R47 as f am(u,m) and in Mathematica as `JacobiAmplitude[u,m]`. Here, m is the modulus representing the square of eccentricity, as before.

This function is the inverse of the elliptic integral of the first kind. To test this on the C47/R47, try an example of $\phi = .1$, $m = .7$. Calculation of elliptic integral of first kind:

[.] [7] [f] [x^2] [.] [1] [f] [F(ϕ,m)] → X: 0.100 08

Calculation of inverse with same modulus:

[.] [7] [f] [x^2] [.] [1] [0] [0] [0] [8] [f] [am(u,m)] → X: 0.100 00

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6 Summary of C47/R47 Elliptic Functions

Table 4 lists the C47/R47 functions this note has described, drawn as the soft keys that reach them on the single page of `Ellipt`. A plain key sits on the menu's first row; an orange one takes the f shift first, and a blue one the g shift. The last three rows are the parameter conversions rather than elliptic functions, and they are on the page because that is where they are needed. Figure 2 is the page itself, for checking the table against.

Table 4 The elliptic functions described here, and where each one sits on the single page of the Ellipt menu.

SOFT KEY	FUNCTION
f am(u,m)	Jacobi Amplitude Function
sn(u,m)	Jacobi sn Function
cn(u,m)	Jacobi cn Function
dn(u,m)	Jacobi dn Function
K(m)	Complete Elliptic Integral of First Kind
f F(ϕ,m)	Incomplete Elliptic Integral of First Kind
E(m)	Complete Elliptic Integral of Second Kind
f E(ϕ,m)	Incomplete Elliptic Integral of Second Kind
Π(n,m)	Elliptic Integral of Third Kind
f Z(ϕ,m)	Jacobi Zeta
g a,b→m	Semi-axes to modulus, $m = 1 - (a/b)^2$ with a the smaller
f k→m	Modulus convention k to m, that is $m = k^2$
f m→k	Modulus convention m to k

In each case except for $\Pi(n, m)$, the u or ϕ parameter is the primary argument, and the m parameter is the square of eccentricity of the ellipse. For $\Pi(n, m)$, n is the characteristic and m is the square of eccentricity.

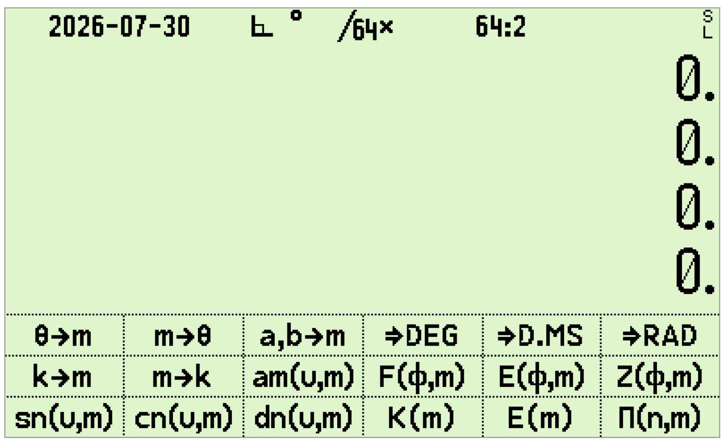

Figure 2 The Ellipt page, which is the whole menu: the unshifted row along the bottom, the f row above it, the g row above that. The tiny S over L at the top right is the simulator's stack-lift indicator; hardware carries the battery gauge there.

The C47 keyboard

Primary function on the key; orange = f (shift once) above; blue = g (shift twice) below.

→I	y^x	x^2	10^x	e^x	α
Σ+	**1/x**	**$\sqrt{x}$**	**LOG**	**LN**	**XEQ**
a b/c	#	.ms	.d	LBL	GTO
|x|	%	π	ASIN	ACOS	ATAN
STO	**RCL**	**R↓**	**SIN**	**COS**	**TAN**
∡	Δ%	$\sqrt[x]{y}$	i	→R	→P

COMPLEX	LASTx	MODE	DISP	UNDO
ENTER	**x⇄y**	**+/−**	**EEX**	**←**
CPX	STK	TRG	EXP	CLR
BST	EQN	ADV	MATX	STAT
↑	**7**	**8**	**9**	**÷**
REGS	HOME	FIN	X.FN	PLOT
SST	BASE	CONV	FLAG	PROB
↓	**4**	**5**	**6**	**×**
FLGS	BITS	CLK	REAL	INTS
	ASN	USER	P.FN	PRINT
f / g	**1**	**2**	**3**	**−**
	KEYS	α.FN	LOOP	I/O
OFF	VIEW	SHOW	PRGM	CAT
EXIT	**0**	**·**	**R/S**	**+**
SNAP	STOPW	INFO	TEST	CNST

The R47 keyboard

The R47 has two physical shift keys: a yellow f and a blue g. Primary on the key; f above, g below.

f (above)	Key	g (below)
i	x^2	→R
$i_{\odot}$	$\sqrt{x}$	→P
$x!$	$1/x$	.ms
$\sqrt[x]{y}$	y^x	.d
10^x	LOG	→I
e^x	LN	#
|x|	STO	∡
%	RCL	Δ%
π	R↓	R↑
USER	DRG	ASN
	f	
	g	
COMPLEX	ENTER	CPX
LASTx	x⇄y	STK
DISP	+/−	TRG
PFX	EEX	EXP
UNDO	←	CLR
α	XEQ	GTO
SIN	7	ASIN
COS	8	ACOS
TAN	9	ATAN
STAT	÷	PLOT
BST	↑	REGS
BASE	4	BITS
INTS	5	REAL
MATX	6	X.FN
EQN	×	ADV
SST	↓	FLGS
PREF	1	KEYS
CONV	2	CLK
FLAG	3	α.FN
PROB	−	FIN
OFF	EXIT	INFO
VIEW	0	I/O
SHOW	·	a b/c
PRGM	R/S	P.FN
CAT	+	CNST

References

[1] Lawden, Derek F, *Elliptic Functions and Applications*, Springer-Verlag, 1989

[2] Olmsted, John M H, *Advanced Calculus*, Prentice-Hall, 1961

[3] Del Grande and Egsgard, *Relations*, Gage Educational Publishing, 1972

[4] StackExchange: *What even are elliptic functions?*. `https://math.stackexchange.com/questions/3383769/what-even-are-elliptic-functions` (Accessed: April 2026).

[5] Cook, J. D., *Clearing up the confusion around Jacobi functions* `https://www.johndcook.com/blog/2018/10/12/jacobi-function-nomenclature/`

[6] StackExchange: *Elliptic integral of the first kind as conformal mapping*. `https://math.stackexchange.com/questions/2341813/elliptic-integral-of-the-first-kind-as-conformal-mapping` (Accessed: April 2026)

[7] StackExchange: *Inverse of elliptic integral of second kind*. `https://math.stackexchange.com/questions/1123360/inverse-of-elliptic-integral-of-second-kind` (Accessed: April 2026)

[8] Wikipedia: Elliptic integral `https://en.wikipedia.org/wiki/Elliptic_integral` (Accessed: April 2026)

[9] C47/R47 Application Note 029, *First Steps*.

AUTHORS

William Gordon Rutherdale	This note
Jaco Mostert	Annex 1, the worked tutorial appended to the published PDF

ANNEX 1: Tutorial to Appnote 23: calculating with ellipses

Below, complimentary to the AN0023 by Will, follows:

1. Explanatory sketches clarifying the variable used by the c47/R47 implementations,
2. Explanatory notes relating to the sketches,
3. Two worked examples, complete with detailed diagrams,
4. 20 problems with solutions, verified on C47/R47, excerpted from Baker, 1890 (see footnote 1 below).

Let me quote Baker from 1890 (see footnote 1 below):

> *It is in the hope of smoothing the road to this interesting and increasingly important branch of Mathematics, and of putting within reach of the English student a tolerably complete outline of the subject, clothed in simple mathematical language and methods, that the present work has been compiled.*
>
> *New or original methods of treatment are not to be looked for. The most that can be expected will be the simplifying of methods and the reduction of them to such as will be intelligible to the average student of Higher Mathematics.*

Funny that in 1890 already, this work was considered mature enough to not want new fashioned treatments ;-) 130 years later the R47/C47 implements still the same math with no new treatments.

R47/C47 X.FN menu ➔ R47/C47 Elliptic / Jacobi functions:

Ellipt	Orthog	$J_y(x)$	$Y_y(x)$	erf	erfc
W_m	W_p	W^{-1}	I_{xyz}	$\beta(x,y)$	LNβ
γ_{xy}	Γ_{xy}	$I\Gamma_p$	$I\Gamma_q$	$\Gamma(x)$	LNΓ

θ→m	m→θ	a,b→m	➔DEG	➔D.MS	➔RAD
k→m	m→k	am(u,m)	F(ϕ,m)	E(ϕ,m)	Z(ϕ,m)
sn(u,m)	cn(u,m)	dn(u,m)	K(m)	E(m)	Π(n,m)

Fig. 1: The worked examples below can be replicated with the R47/C47 Ellipt *menu functions above:*

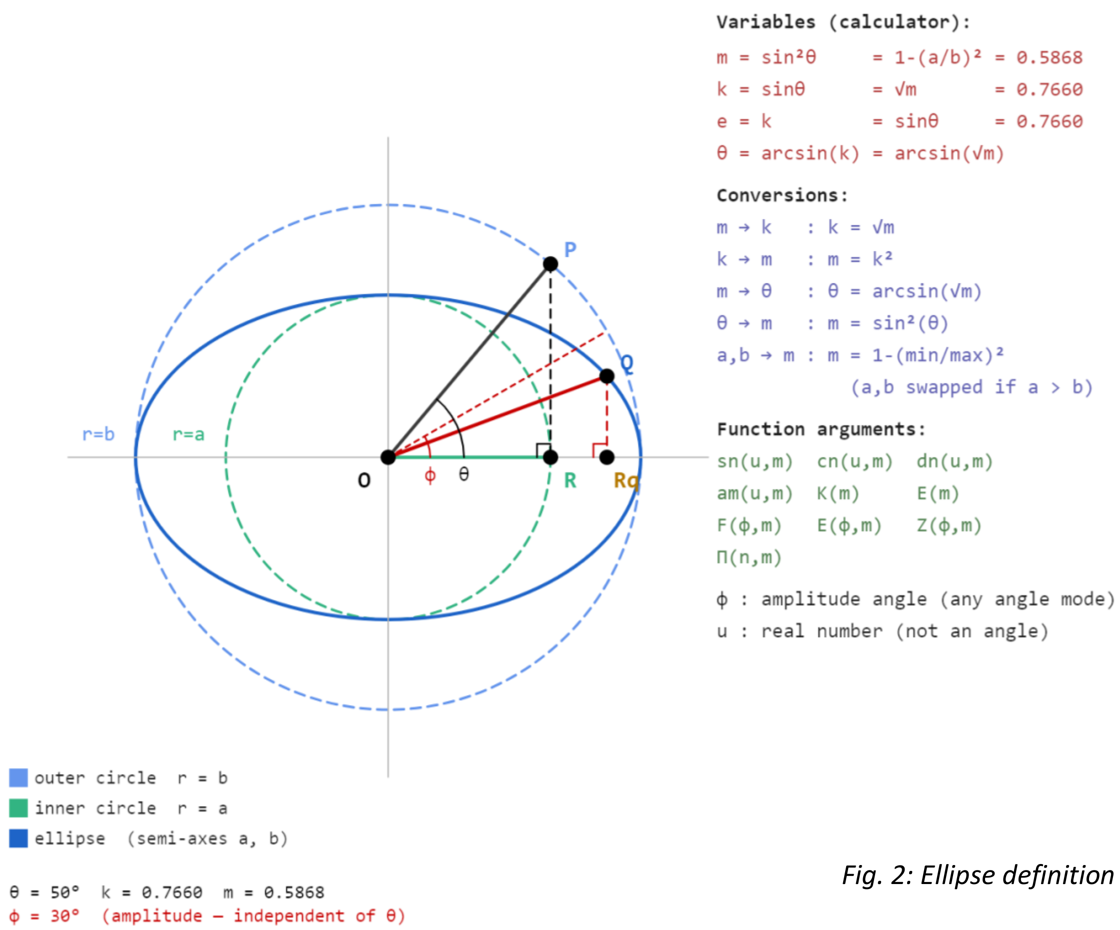

Fig. 2: Ellipse definition & command summary

The ellipse above is fully described by its two semi-axes a and b. From these the calculator parameter $m = 1-(a/b)^2$ and modulus $k = \sqrt{m}$ follow directly.

The eccentricity angle $\theta = \arcsin(k)$ ties the ellipse shape to the function parameters. With P on the outer circle at angle θ, the vertical from P drops to R on the major axis and OR = a, giving $\cos\theta = a/b$. If a and b are known, $m = 1-(a/b)^2$ follows directly. If the ellipse is given as a drawing, θ is the angle to measure to get m without needing to know a and b explicitly.

For a point Q on the ellipse, the amplitude angle ϕ gives the coordinates of Q through the ellipse equations $x = b\cdot\cos\phi$ and $y = a\cdot\sin\phi$. This is the critical distinction the diagram is meant to convey: ϕ is not the geometric angle QOR visible in the figure, which would be $\arctan(y/x)$ evaluated at Q.

The two different angles θ and ϕ, at the same point, are most easily confused.

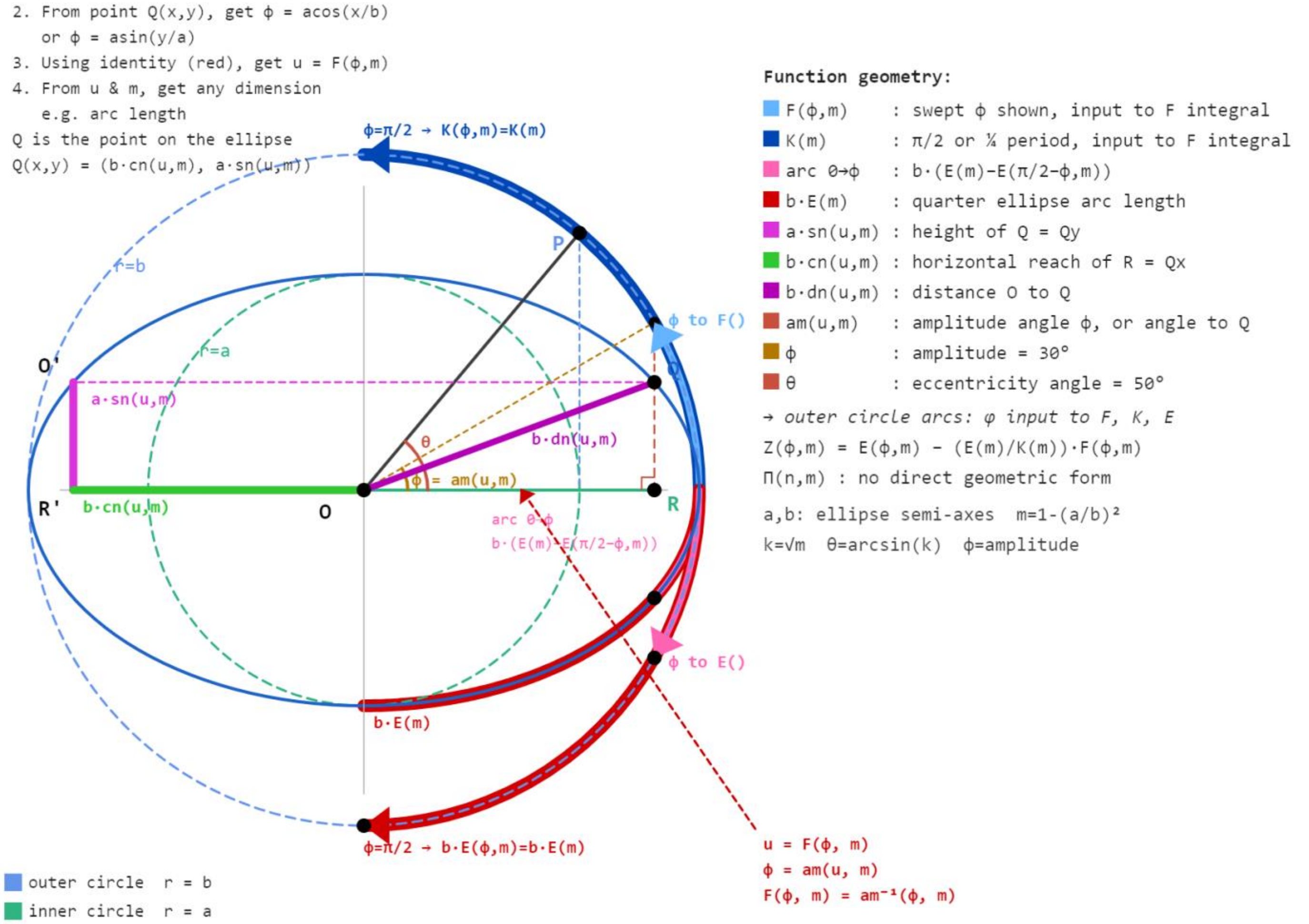

Fig. 3: Elliptical Function Summary and clarification

Once m and ϕ are known, u = F(ϕ,m) is the single quantity that unlocks all Jacobi functions. F is the inverse of the amplitude function: ϕ = am(u,m), so u is simply the argument whose amplitude is ϕ. The diagram shows this geometrically – the blue arc sweeps ϕ as input to F, and K(m) = F(π/2,m) is the same function evaluated at its quarter-period input.

The arc length along the ellipse from 0 to Q is b·(E(m) − E(π/2−ϕ,m)), not b·E(ϕ,m). The distinction matters in practice: E(ϕ,m) alone does not give arc length except at the quarter period where it equals b·E(m) exactly. The correction term E(π/2−ϕ,m) accounts for the asymmetry of the integrand between the two ends of the quarter arc.

The segments b·cn(u,m), a·sn(u,m) and b·dn(u,m) form a right triangle OQR, so once u and m are known all three dimensions of Q are available simultaneously.

A single call to F gives u, and from u the full geometry of the ellipse follows in one step.

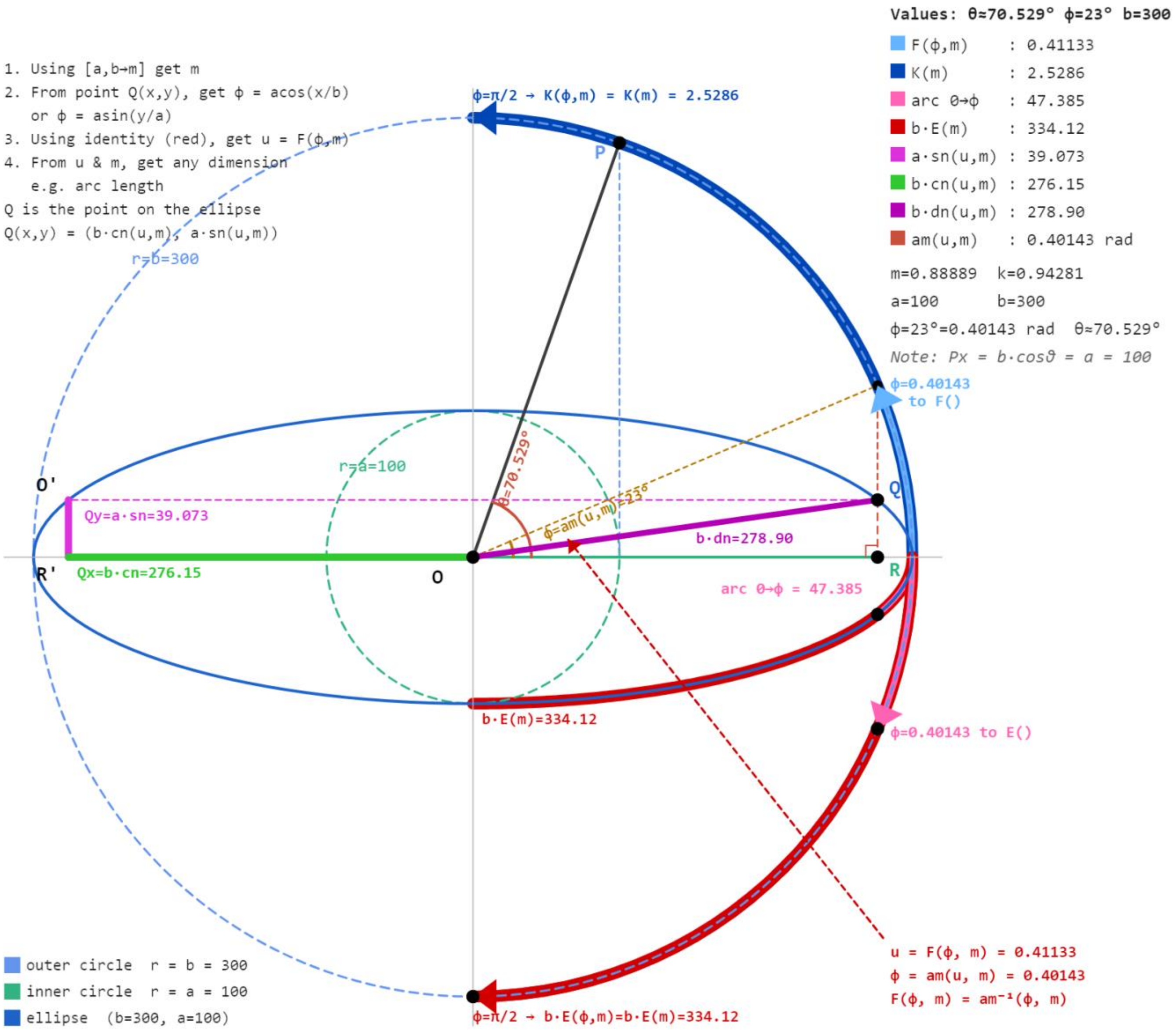

Fig. 4: Elliptical Numerical Solutions 1

Above, the highly eccentric ellipse with a=100 and b=300 which gives m=0.889 and k=0.943. At this eccentricity the elliptic functions behave quite differently from a near-circular case, and the geometry is more visible precisely because the ellipse is so elongated.

The instruction block top left gives the workflow in four steps: get m from a and b, get ϕ from the coordinates of Q, get u from F(ϕ,m), then read off any dimension. Everything flows from m and ϕ.

The blue arc shows ϕ=23° being fed into F. The output u=0.41133 is neither an angle nor a length and does not appear on the drawing. It is the argument to all the Jacobi functions. From u and m the calculator returns b·cn=276.15, a·sn=39.07 and b·dn=278.90 as the three sides of the right triangle around the point on the ellipse, i.e. OQRq. K(m)=2.529 is F evaluated at ϕ=π/2, the quarter period input.

The arc OQ along the ellipse is 47.385, obtained as b·(E(m)−E(67°,m)) where 67°=90°−ϕ. The full quarter arc b·E(m)=334.12 is considerably longer than a less eccentric ellipse would give, reflecting how much the curve stretches near the major axis when k approaches 1.

The red identity block (bottom right) states u=F(ϕ,m) and ϕ=am(u,m) as inverses.

To solve the ellipse, starting with a coordinate, arc length, or angle, requires finding ϕ, then u.

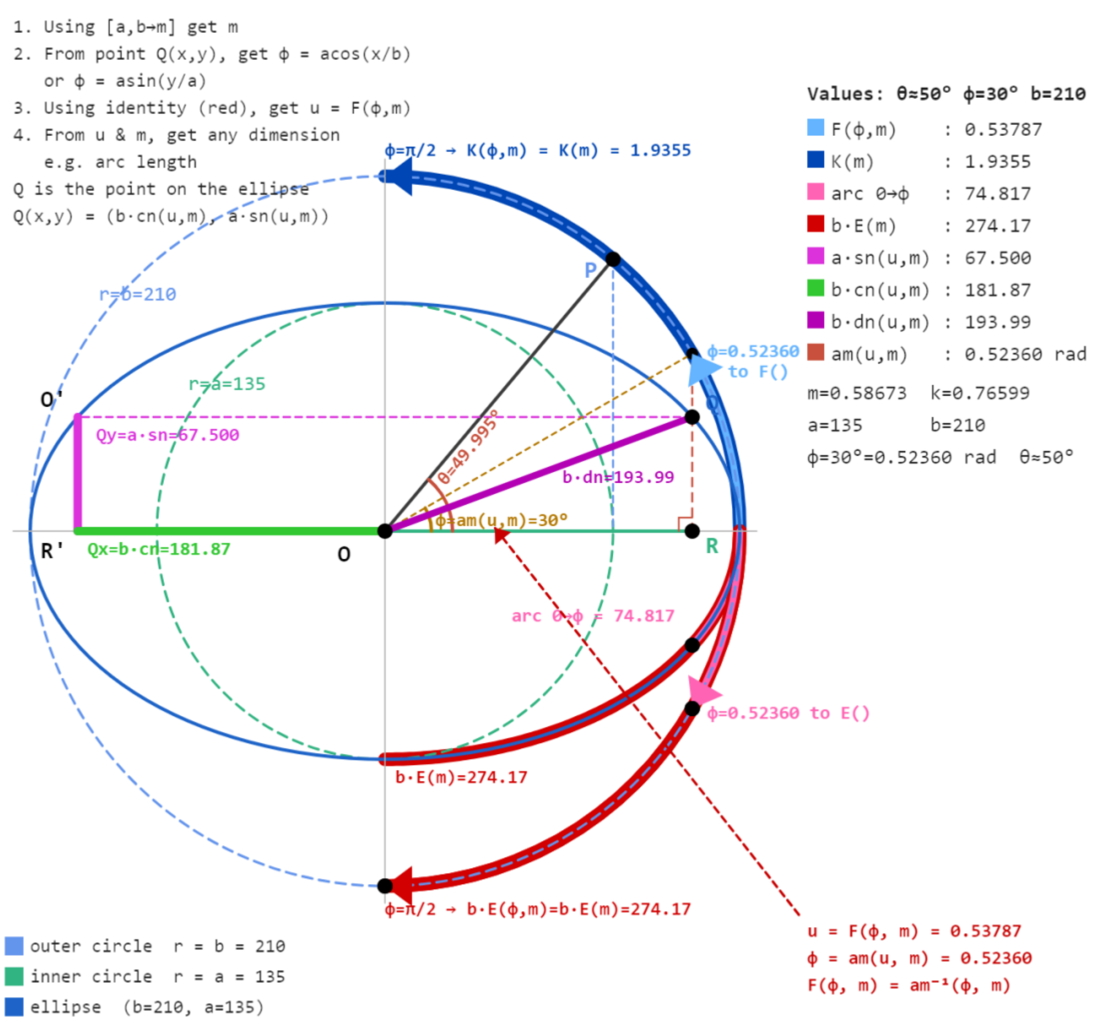

Fig. 5: Elliptical Numerical Solutions 2

Second exercise left to user, with no comments

Examples from Baker[1],

1. p.106: Let k= sin θ= sin 19°30′. Required K. K= 1.615101 (*ed. Printing error. 3 digits correct in the book; log K was given correctly as log K=0.208852, implying their log-book lookup was wrong*).

2. p.108: Given k= sin 75°. Find K. K= 2.768064 Ans.

3. p.109: Given k= sin 45°. Find K. K= 1.8540747 Ans.

4. p.110: Given θ= 63°30′. Find K. Ans. log K= 0.3539686

5. p.110: Given θ= 34°30′. Find K. Ans. K= 1.72627

6. p.111: Let ϕ = 30° k= sin 45°. Find u. u =0.535623, Ans.

7. P.113: Given k= sin 75°, tan ϕ = √(2/√3). To find F(k, ϕ). F(k, ϕ) = 0.9226878. Ans.

8. p.117: Given ϕ = 30°, k= sin 89°. Find u. F(k, ϕ) = 0.549297. Ans.

9. p.118: Given = 79°, k = 0.25882. Find u. Ans. u = 0.39947. (*ed. Printing error: 1.39947*).

10. p.118: Given = 37°, k = 0.86603. Find u. Ans. u =0.68141.

11. p.119: Given u = 1.368407, θ = 38°. Find ϕ. ϕ = 72°. Ans.

12. p.122: Given u = 2.41569, θ = 80°. Find ϕ. Ans. ϕ = 82°

13. p.122: Given u = 1.62530, k = 1/2 . Find ϕ. Ans. ϕ = 87°

14. p.123: Given k= 0.9327, ϕ = 80°. Find E(k, ϕ). E(k, ϕ) = 1.0562 (*Ed. Incorrect, it is 1.06349*)

15. p.123: Given k= sin 75°, tan ϕ = √(2/√3). To find E(k, ϕ). E(k, ϕ) = 0.7387196. Ans.

16. p.127: Given k = sin 75°. Find E(k, π). E(k, π) = 1.076405. Ans.

17. p.127: Given k =sin 30°, ϕ = 81°. Find E(k, 0). Ans. E(k, 0) =1.33124.

18. p.127: Find E(sin 80°, ϕ = 55°). Ans. 0.82417.

19. p.127: Find E(sin 27°, ϕ = π /2). Ans. 1.48642.

20. p.127: Find E(sin 19°, ϕ = 27°). Ans. 0.46946.

[1] *Baker Arthur L, Elliptic Functions An Elementary Text-Book for Students of Mathematics, New York John Wiley and Sons, 1890, re-released Project Gutenberg, January 25, 2010 [eBook #31076]*